Advait Nair

832-493-4529 advaitnair.work@gmail.com linkedin.com/in/advaitnair10
Austin, TX advaitnaireng.github.io/advaitnair U.S. Citizen

Professional Summary

First-year Civil Engineering student at UT Austin seeking a civil/structural engineering internship, with project experience in building site design, stormwater drainage, and structural testing. Designed a 49,486 sq ft teaching building and its 4.83 ft3/s stormwater drainage system, and measured a race-car frame's torsional stiffness to validate a structural design. U.S. Citizen, valid Texas driver's license.

Education

UT Austin Austin, TX
B.S. in Civil Engineering Expected May 2030
Relevant coursework: Civil Engineering Systems; Sequences, Series, and Multivariable Calculus; Principles of Chemistry I

Experience

Longhorn Racing Electric - UT Austin Formula SAE Electric Austin, TX
Frame Design Engineer Sep. 2026 – Present
- Validated a race-car frame's structural stiffness: built a Python sizing calculator that set the torsion jig's fixture heights, applied load, and deflection targets, ending roughly a month of team design deliberation
- Building a torsional-rigidity test jig to measure a race-car frame's structural torsional stiffness by applied load and beam deflection, sizing fixtures against floor-clearance constraints
- Optimizing frame stiffness per unit mass in Ansys by evaluating chassis node configurations in the cockpit area, increasing stiffness by roughly 200 Nm/degree for a 1.91 lb weight increase

ShorelineAI Remote
Engineering Intern May 2024 – Aug. 2024
- Created a CAD model where no source model existed by reverse engineering a wind-direction sensor in Onshape from photographs and physical measurements
- Delivered a 3D-printable bolt-on mount for a steel crossarm that blocks rotation without drilling, selecting ASA for field conditions and handing the CAD assembly and drawings to the hardware team

Korgel Group, UT Austin Austin, TX
CAD Design Contributor Oct. 2026 – Present
- Building SolidWorks CAD models of shadow masks that define selective gold-deposition patterns on glass substrates for solar-cell fabrication via evaporation deposition

Projects

Civil Engineering Teaching Center - CE 301 Design Project *Ongoing project*
- Designed a three-story, 49,486 sq ft teaching building for UT Austin, producing AutoCAD site and floor plans for an L-shaped layout around a preserved heritage oak, exceeding the 37,753 sq ft minimum area requirement
- Set building placement and grading from a field survey of existing sidewalks, fire hydrants, storm inlets, and topography
- Specified concrete and cast iron pipes by Manning capacity and slope, with a headwall and erosion protection at the Waller Creek outlet
- Delivered a gravity stormwater system sized for the 100-year, 15-minute storm (NOAA Atlas 14, 10.2 in/hr): 5 inlets and 7 Manning-sized pipes carrying 4.83 ft3/s to Waller Creek, every pipe capacity exceeding demand; quantified runoff increase at 3.32 ft3/s; authored the drainage design report

Beam Bending Lab - CE 301 Civil Engineering Systems
- Determined beam stiffness from measured midspan deflection of pine, Douglas fir, aluminum, and steel beams over a 4-ft span in upright and flat orientations, calculating moment of inertia and Young's modulus (pine 1.89×10^6 psi, Douglas fir 2.53×10^6 psi); upright beams were stiffer because height is cubed in the moment-of-inertia equation
- Selected the lowest-cost beam meeting the serviceability limit: an upright pine 2×6 for a 250-lb load over a 10-ft span, with predicted deflection 0.229 in. against a 0.500-in. limit at $6

Wind Turbine Blade Material Selection - Independent R Project
- Built an R multi-criteria model comparing aluminum alloy, fiberglass, and carbon composites for wind turbine blades, scoring specific stiffness and specific strength per unit weight
- Weighted density, stiffness, fatigue, and cost across three design-priority scenarios; carbon composite led on performance (0.90) and balance (0.85) while fiberglass won when cost dominated (0.67 vs 0.60)

Skills and Licenses

Design & CAD: AutoCAD, SolidWorks, Ansys, Onshape
Programming & Data: Python, R, Excel, Microsoft Word, Outlook
Hands-on Fabrication: Basic TIG welding, 3D printing
License: Valid Texas driver's license

Leadership & Activities

Technology Student Association - Chapter President *2022 – 2026*
- Led 120+ members through 90+ meetings and managed up to $13,000 in chapter funds; placed 3rd in Debating Technical Issues at the International Conference

Seven Lakes Speech and Debate - Extemporaneous Speaking Captain *2022 – 2026*
- State finalist (4th in Impromptu of about 1,000 competitors); national qualifier; 40th in Texas Congressional Debate; NSDA Premier Distinction (top 3% of approximately 141,132 student members)